



AKGEC SKILLS FOUNDATION

Training Program Details

CENTRE NAME- BOSCH AUTOMOTIVE CENTRE OF EXCELLANCE

Training/Course Title:	Industrial Training Program on Automotive Technicians
Target Audience:	ITI/Diploma Students
Fee:	6000/-
Duration:	160 Hrs.
Module	• Fundamental principles of automotive engineering, including vehicle architecture, subsystem interaction, and system-level functionality across powertrain, chassis, electrical, and safety domains. • Structural configuration and functional analysis of chassis systems, suspension mechanisms, axle assemblies, and steering kinematics with emphasis on vehicle dynamics. • Thermodynamic and mechanical operation of internal combustion engines, including combustion cycles, component functions, lubrication, and cooling systems. • Transmission engineering covering clutch mechanisms, manual and automatic gearbox architecture, differential operation, and complete driveline power flow. • Automotive electrical and electronic systems including battery technology, starter and charging systems, wiring harness architecture, and sensor–actuator integration. • Engine management systems with ECU architecture, control strategies, diagnostic protocols, sensor characterization, actuator response analysis, and fault code interpretation. • Brake engineering principles including hydraulic braking fundamentals, disc–drum systems, and safety-critical system behaviour. • Air-conditioning and thermal management systems including refrigeration cycle analysis, refrigerant properties, compressor operation, system diagnostics, and servicing procedures. • Tool engineering and workshop technology involving correct tool classification, mechanical principles, equipment operation (including alignment machines, test benches), and workshop safety protocols.

	Automotive safety engineering covering crash dynamics, occupant protection systems, crash sensors, airbag deployment mechanisms, diagnostic inspection, and compliance with safety standards.	
Outcome:	- Ability to interpret and analyse the operational principles, architecture, and interdependency of major automotive systems using engineering fundamentals. - Competence in performing mechanical diagnostic procedures including engine disassembly/assembly, transmission inspection, brake evaluation, and AC system servicing. - Proficiency in understanding, troubleshooting, and maintaining automotive electrical systems, including charging, starting, wiring harnesses, and sensor-based subsystems. - Capability to operate advanced diagnostic tools (e.g., KTS scanners), interpret ECU fault codes, evaluate system performance, and execute systematic troubleshooting. - Aptitude to assess safety-critical automotive systems, including braking and stability systems, crash sensors, and occupant protection modules. - Technical skill in applying workshop practices, tool handling, mechanical measurement, and adherence to industry safety protocols for maintenance and repair operations. - Readiness to perform professional automotive inspection, maintenance, and diagnostic tasks aligned with industrial standards and service requirements.	
Certification:	The participants, on successful completion of above trainings, will receive joint certification by BOSCH India & AKGEC.	
Program Coordinator:	Name: Mr. Jitendra Sirohi	Contact number: 9999972391
Payment Mode:		